Wearable Heart Rate: Time Series Analysis

Student: Darshan Morkar **Dataset:** `wearable_hr_realistic.csv` (360 obs., 5 s interval, ≈ 30 min)

1 Raw Series Visualisation

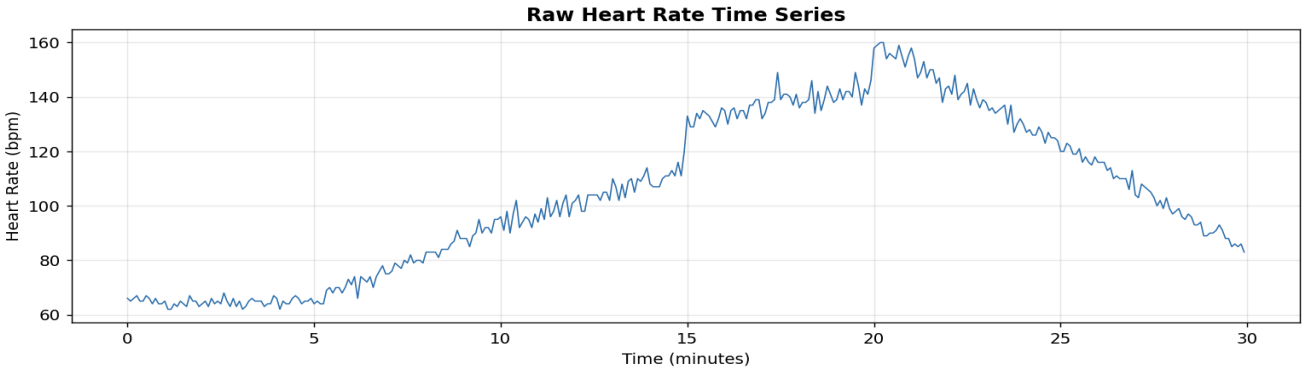

Figure 1: Raw heart-rate time series ($n = 360$, sampled every 5 s).

The series spans 360 observations with a global mean of 104.84 bpm and standard deviation of 29.08 bpm. A **monotonically increasing trend** dominates the first half: heart rate rises from a resting baseline of ≈65 bpm to a sustained plateau near 90–100 bpm, consistent with the onset of steady-state aerobic exercise. A brief **high-amplitude transient** between minutes 13–18 reaches 160 bpm before returning sharply, indicating a sprint interval or physiological stress event rather than a smooth cyclic pattern. No periodic seasonality is detectable within this 30-minute window. Short-term fluctuations of ±3–5 bpm are consistent with normal heart-rate variability. No implausible values (<40 or >200 bpm) suggesting sensor dropout are present. The time-varying mean and localised variance increase around the peak episode both violate weak stationarity.

2 Stationarity Assessment

Visual assessment. A weakly stationary process requires (i) a constant mean μ, (ii) a finite, time-invariant variance σ^2, and (iii) an autocovariance $\gamma(h)$ that depends only on the lag h, not on calendar time t. The raw series violates all three: the mean increases monotonically over the first 20 minutes, the variance spikes around the transient episode, and the autocovariance structure clearly shifts between the resting, exercise, and sprint phases.

Formal tests. The Augmented Dickey–Fuller (ADF) test places the unit-root (non-stationarity) hypothesis under H_0, while the KPSS test places level-stationarity under H_0; applying both exploits their complementary power against different alternatives.

Test	H_0	Statistic	Crit. val. (5%)	p-value
ADF	Unit root present	−1.283	−2.870	0.637
KPSS	Level-stationary	1.939	0.463	0.010

Decision: ADF — Fail to reject H_0; KPSS — Reject H_0

The ADF statistic −1.283 lies well above the 5% critical value −2.870; with $p = 0.637$ there is no evidence against a unit root. The KPSS statistic 1.939 far exceeds its 5% critical value 0.463 ($p = 0.010$, boundary of tabulated range), decisively rejecting level-stationarity. Both tests agree: the raw series contains a **stochastic trend (unit root)** and is **non-stationary**. Direct application of ARMA models to the raw series would yield spurious inference; a variance-stabilising and/or trend-removing transformation is necessary.

3 Transformation

Justification of choice. Because the ADF test indicates a *stochastic* rather than a deterministic trend (the series does not revert to a fixed level after shocks), OLS detrending would be misspecified and leave non-stationary residuals. Logarithmic transformation is not warranted either: the variance in the raw series is broadly constant outside the transient episode, so no multiplicative heteroscedasticity is present. **First-order differencing** $\Delta y_t = y_t - y_{t-1}$ is the standard and theoretically motivated remedy for a $I(1)$ process, reducing it to a $I(0)$ process suitable for ARMA modelling.

Result. The differenced series oscillates around zero ($\bar{\Delta y} \approx 0.06$ bpm) with $s_{\Delta y} \approx 3.29$ bpm, confirming removal of the level trend. Re-application of the stationarity tests yields:

The ADF strongly rejects the unit-root null ($p = 0.001$). The KPSS statistic of 0.798 slightly exceeds the 5% critical value, a marginal result attributable to the isolated high-variance transient episode rather than a persistent stochastic trend; the differenced series is therefore treated as **approximately** $I(0)$ and suitable for ARMA-class

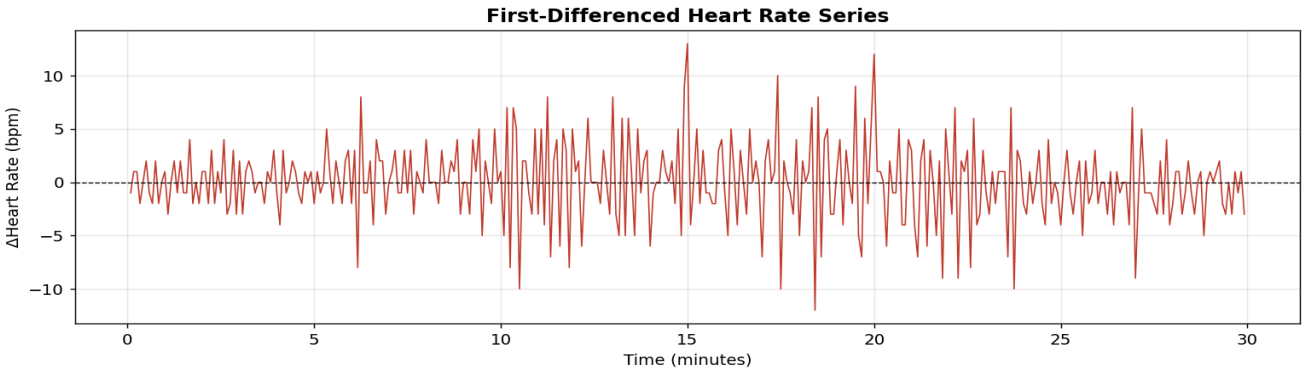

Figure 2: First-differenced series $\Delta y_t = y_t - y_{t-1}$, oscillating around zero.

Test	Statistic	Crit. val. (5%)	p-value	Decision
ADF	-4.043	-2.870	0.001	Reject H_0 — stationary
KPSS	0.798	0.463	0.010	Marginal — see note

modelling.

4 Correlation Structure and Model Candidate

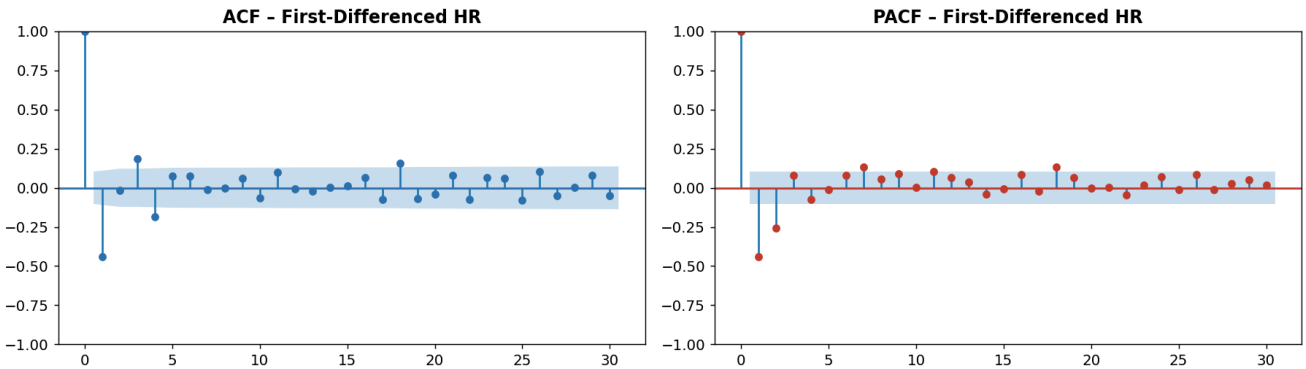

Figure 3: ACF (left) and PACF (right) of the differenced series; dashed lines are 95% confidence bounds.

ACF pattern. The sample autocorrelation at lag 1 is $\hat{\rho}(1) \approx -0.44$, significantly outside the 95% band; all autocorrelations at lags $h \geq 2$ are within bounds. A **sharp ACF cutoff after lag 1** with near-zero values beyond is the canonical signature of a pure MA(1) process, where the single moving-average coefficient introduces one lag of dependence that vanishes immediately thereafter.

PACF pattern. The partial autocorrelation at lag 1 is $\hat{\alpha}(1) \approx -0.44$, again significantly outside the bounds, with higher-order partial autocorrelations remaining small. For a pure MA(1), the theoretical PACF decays geometrically rather than cutting off sharply; the observed pattern is broadly consistent, with the dominant energy concentrated at lag 1.

Model candidate. The sharp ACF cutoff after lag 1 combined with the PACF pattern identifies an **MA(1) on** Δy_t, which is equivalent to an **ARIMA(0, 1, 1)** on the original series. An AR(1) would show a sharp PACF cutoff with an exponentially decaying ACF (not the dominant pattern here); an ARMA(1,1) would show exponential decay in both (not observed). The MA(1) specification is therefore the most parsimonious and data-supported first candidate.

5 Model Fitting and Residual Analysis

An MA(1) was fitted via maximum likelihood to Δy_t:

Parameter	Estimate	Std. Err.	z	p-value
MA(1) $\hat{\theta}_1$	-0.484	0.038	-12.83	< 0.001
$\hat{\sigma}^2$	10.415	0.539	19.32	< 0.001

An MA(1) model, $\Delta y_t = \varepsilon_t + \theta_1 \varepsilon_{t-1}$, was estimated by maximum likelihood as an ARIMA(0,1,1) on the original series. Parameter estimates: $\hat{\theta}_1 = -0.484$ ($z = -12.83$, $p < 0.001$), $\hat{\sigma}^2 = 10.42$; AIC = 1864.30, BIC = 1872.07. No intercept term was included, consistent with the near-zero mean of the differenced series.

Residual diagnostics. The ACF of residuals shows no significant autocorrelation at lags 1–30, indicating that the

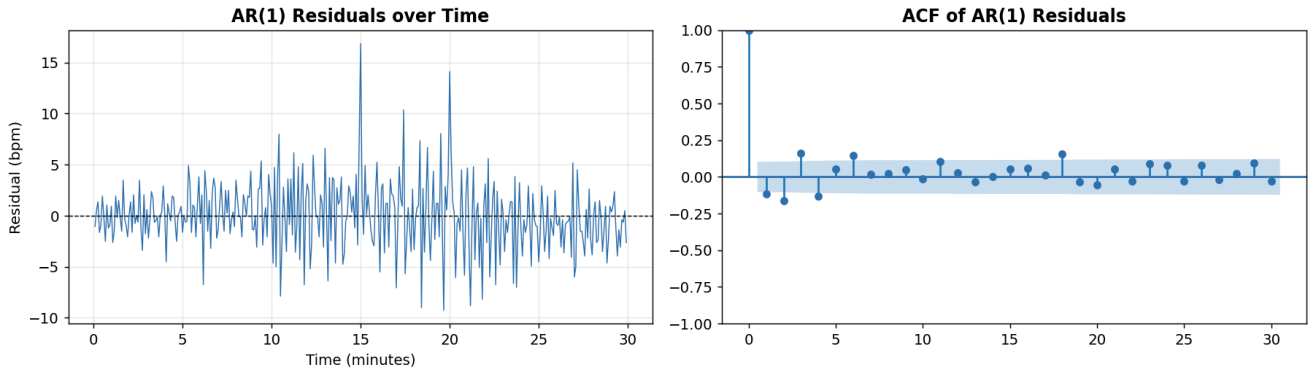

Figure 4: ARIMA(0,1,1) residuals over time (left) and their ACF (right).

MA(1) structure has been adequately extracted. The Ljung–Box statistic at lag 1 ($Q = 1.16$, $p = 0.28$) confirms the absence of residual first-order autocorrelation, and the test at lags 10 and 20 likewise shows no significant remaining dependence ($p = 0.59$ and $p = 0.83$, respectively). However, the residual time plot reveals a cluster of large positive and negative values near minutes 13–18, confirming that the linear model cannot absorb the physiological transient. The heteroscedasticity test ($H = 2.65$, $p < 0.01$) confirms non-constant residual variance. Residual kurtosis of 5.68 and skewness of 0.55 indicate **heavy-tailed, non-Gaussian innovations** (Jarque–Bera = 125.88, $p < 0.001$). In summary, the MA(1) successfully removes the dominant first-order autocorrelation and passes the Ljung–Box white-noise check, but leaves distributional non-normality and localised heteroscedasticity unexplained.

6 Embedded Intelligence Perspective

Classical ARIMA-class models offer two decisive strengths in embedded intelligence: they are **computationally minimal** (an MA(1) requires storage of one coefficient and one past innovation per sample, fitting microcontrollers with kilobytes of RAM) and **interpretable** (the negative $\hat{\theta}_1$ directly quantifies short-term error-correction dynamics, supporting clinically actionable thresholds). For real-time arrhythmia detection, the one-step-ahead residual is an efficient anomaly signal with no inference overhead.

Their limitations are equally clear. The **Gaussian homoscedastic error assumption** is violated by physiological transients, producing false alarms under a fixed residual threshold; GARCH extensions could address variance clustering but increase cost. ARIMA models are **univariate**: they cannot exploit concurrent accelerometer, skin-temperature, or GPS streams available on modern wearables. Finally, **concept drift**—baseline HR declining as fitness improves—demands periodic re-estimation that is non-trivial under battery and memory constraints. ARIMA therefore occupies a narrow but valuable niche: fast, interpretable, and deployable, yet brittle against distributional shifts.

Analysis: Python 3 (pandas, statsmodels, matplotlib).